Department of Physics
IIT Hyderabad

EP-2108 (Special Relativity)
Time: 1 hour, Total Marks: 50

1. An event occurs in frame S at a point A. A second event occurs at point B in the same frame $0.01\,\mu s$ later. The distance between A and B as measured by an observer in frame S is 5 meters. As viewed the same events by an observer in another frame S', the two events occur simultaneously but at a distance of 4 meters apart. Calculate the velocity of the frame S' with respect to the frame S. If possible calculate the velocity of a frame S'' where the two events occur at the same place. Give full justification for your answer. [15 marks]

2. Let us consider an inertial frame S' that moves with a velocity

$$\vec{v} = \beta c(\cos\phi\,\hat{i} + \sin\phi\,\hat{j})$$

with respect to frame S. If the origins of two frames coincide with each other at $t = t' = 0$ and their axes are parallel to each other, then find the Lorentz matrix Λ for the transformation

$$(x^0, x, y, z) \to (x'^0, x', y', z')$$

where $x^0 = ict$. [10 marks]

Here

$$\beta = \frac{|\vec{v}|}{c} \qquad \text{and} \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

3. A ρ-meson of mass $760\,\mathrm{MeV}/c^2$ decays at rest into two π-mesons of mass $140\,\mathrm{MeV}/c^2$ each. What is the relative velocity of the π-mesons with respect to each other. [10 marks]

4. A particle of rest mass m_0 and speed v collides and sticks to a stationary particle of mass M. What is the final speed of the composite particle. [5 marks]

5. A circular lamina moves with its plane parallel to the X-Y plane of a reference frame S at rest. Assuming its motion to be along the X-axis, calculate the velocity at which its surface area would appear to be reduced to half to an observer in frame S. [10 marks]

— *paper ends* —

The solutions are discussed in the class (SR_13)